

SAI VENKATA ADITHYA CHALLA

Los Angeles, CA, 90007

+1-213-477-9905 | adithyachalla12@gmail.com | <https://www.linkedin.com/in/adithyachalla/>
<https://challaportfolio.vercel.app/>

EDUCATION

University of Southern California

Aug 2024 - May 2026

Master's, Computer Science (GPA: 3.75/4.0)

- **Coursework:** Applied NLP, Database Systems, Information Retrieval, Machine Learning for Data Science, Web Technologies, Deep Learning, Analysis of Algorithms, Software Engineering

GITAM University

Jun 2019 - Apr 2023

Bachelor of Technology, Computer Science (GPA: 9.15/10)

- **Coursework:** Foundations of AI, Data Warehouse & Mining, Object Oriented Programming, Operating Systems, Agile Software Development

TECHNICAL SKILLS

- **Languages & Frameworks:** C, Java, Python, R, PySpark, KQL, MySQL, HTML, CSS, JavaScript, Flask, React, Node.js
- **Cloud & Data Engineering:** AWS S3, AWS EC2, AWS Lambda, AWS Quicksight, Azure Data Factory, Azure Synapse Analytics, Microsoft Fabric
- **Data Analysis & Visualization:** Numpy, Pandas, Matplotlib, Seaborn, Power BI, Power Automate, Tableau, D3.js
- **ML/AI Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, NLTK, OpenCV, Streamlit, HuggingFace, Agent Development Kit (ADK), FastAPI, Transformers
- **DevOps & Tools:** Git, GitHub, Docker, REST APIs, CI/CD

EXPERIENCE

Vosyn | Machine Learning Engineer Intern

Feb 2026 - May 2026

- Owned the reliability, evaluation, and deployment of two production speech AI services (API Gateway, Segmentation, Transcription, Speaker Diarization, and Comet), improving pipeline stability by 30% through Docker optimization, Cloud Run hardening, and standardized runtime configuration
- Diagnosed and resolved transcription quality issues across WhisperX and NVIDIA NeMo pipelines, addressing timestamp drift, segmentation mismatches, and speaker attribution errors to improve transcription alignment and reduce speaker-diarization mismatches by ~22%
- Built CI-driven validation, deployment guardrails, and rollback-ready CI/CD workflows with environment-specific configurations, reducing debugging and regression isolation time by ~25% while enabling more reliable ML service deployments

MAQ Software | Software Engineer 1

Aug 2023 - Jun 2024

- Designed and developed full-stack analytics solutions using React, Power BI Embedded, Azure Data Factory, Databricks, ADLS Gen2, Azure SQL, and KQL, delivering interactive customer-facing dashboards for SLA tracking, ticket aging, escalation trends, and operational monitoring.
- Automated reporting workflows using Power Automate and Azure services, eliminating approximately 3 hours of daily manual effort and improving the timeliness and accuracy of operational reporting
- Developed self-service analytics solutions by integrating incident, telemetry, and access-management datasets using KQL and Azure Data Explorer, reducing engineering support requests by 40% and improving analyst productivity
- Conducted root-cause and trend analysis on operational data to identify recurring escalation drivers and service bottlenecks, providing actionable insights that improved issue resolution efficiency and service reliability

iAmX | AI/ML Intern

Apr 2022 - May 2023

- Developed machine learning models using Python and Scikit-learn to analyze customer purchasing behavior, forecast product demand, and identify sales trends from retail datasets
- Built product recommendation and item-matching solutions using TF-IDF, cosine similarity, and collaborative filtering techniques to improve product discovery and cross-selling opportunities
- Performed feature engineering, model training, hyperparameter tuning, and evaluation to improve predictive performance across customer segmentation and recommendation use cases

PROJECTS

Multi-Agent AI Career Guidance System

- Implemented a multi-agent system architecture using Google ADK and Gemini LLMs, supporting parallel and sequential agent orchestration
- Developed tool-augmented LLM agents integrating external search APIs and currency conversion utilities for structured data retrieval and normalization
- Designed a modular data processing pipeline with parallel execution, aggregation layers, and standardized output schemas for downstream consumption

AI-Powered Roommate Discovery Platform

- Developed RoomSync, an AI-powered roommate discovery platform, using a scalable client-server architecture to support intelligent roommate matching and housing workflows
- Built and optimized user and lister profile systems that captured lifestyle preferences, housing expectations, and behavioral data for AI-driven compatibility recommendations
- Implemented real-time chat and connection features that enabled users to communicate for roommate coordination, housing discussions improving user engagement and platform interaction

Knowledge Conflict Resolution System

- Built an interpretable multimodal knowledge conflict resolution system using CLIP, BLIP, and semantic similarity metrics to detect contradictions between text and images
- Trained a CNN-BiLSTM text backbone on 100K Fakeddit samples, achieving 0.83 accuracy with stable generalization and class-balanced performance
- Conducted evidence-grounded claim verification on the LIAR benchmark using BERT/RoBERTa cross-encoders, reducing adjacent label confusion and improving calibration 0.18-0.07 to inform multimodal decision logic